

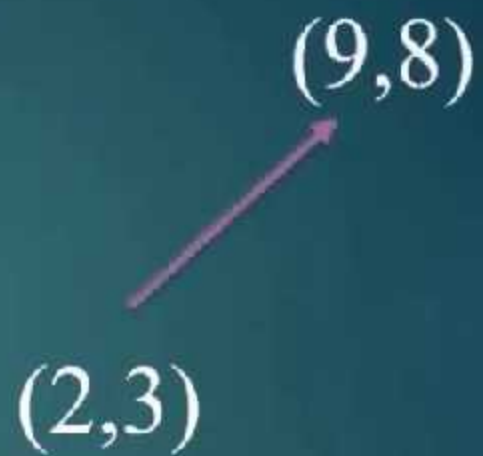
Vectors

2

$(2,3)$

Vectors

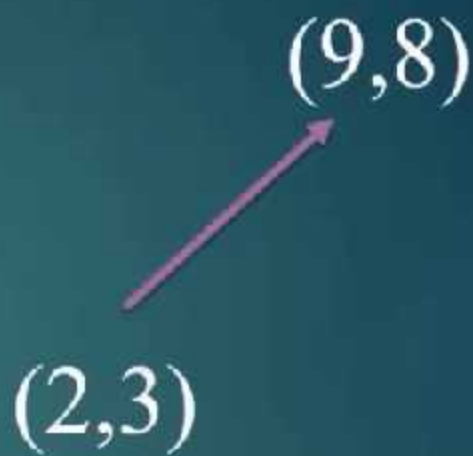
2



Vectors

$$V = (9,8) - (2,3)$$

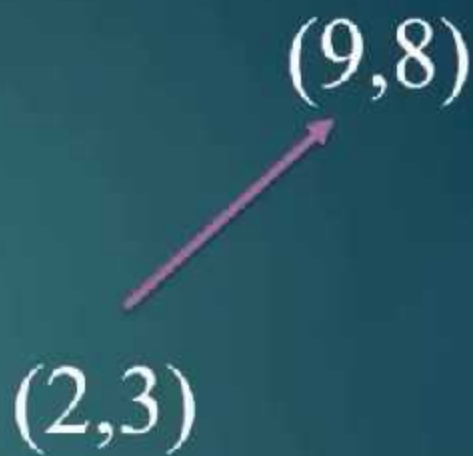
$$V = (7,5)$$



Vectors

$$V = (9,8) - (2,3)$$

$$V = (7,5)$$



Cartesian Form

$$V = 7i + 5j$$

Operations on Vectors

Operations on Vectors

3

► 1- Additions:

$$r = 2i + 3j$$

$$S = 5i + 7j$$

Operations on Vectors

3

► 1- Additions:

$$r = 2i + 3j$$

$$S = 5i + 7j$$

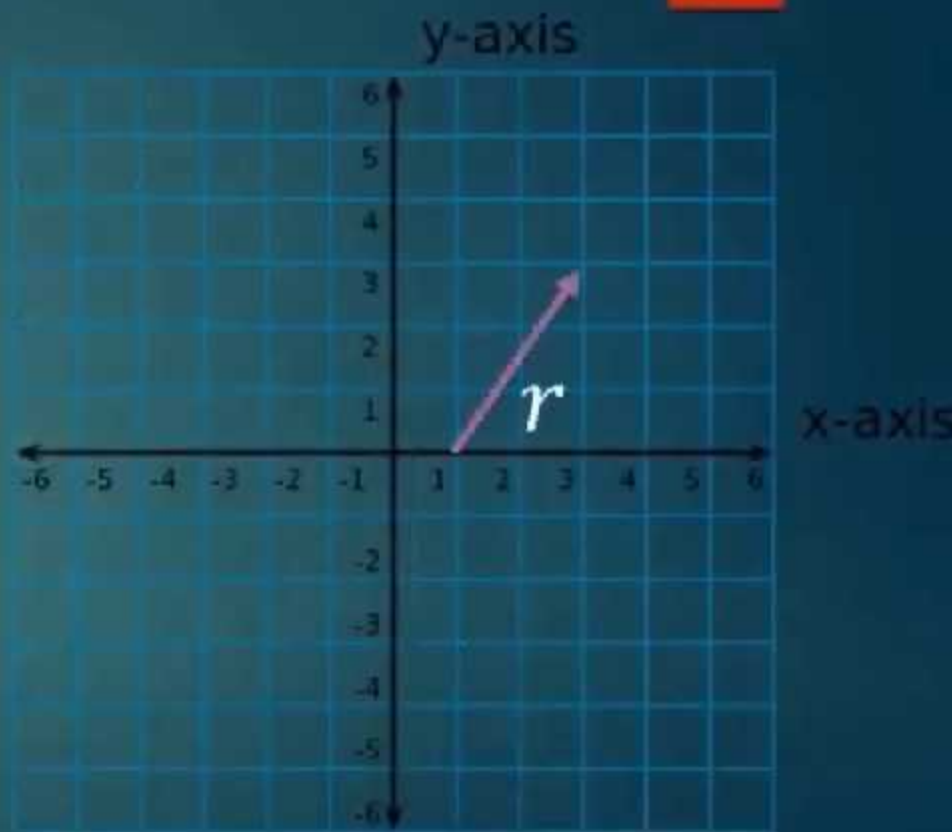
$$r + s = 7i + 10j$$

Operations on Vectors

4

► 1- Additions:

$$r = 2i + 3j$$



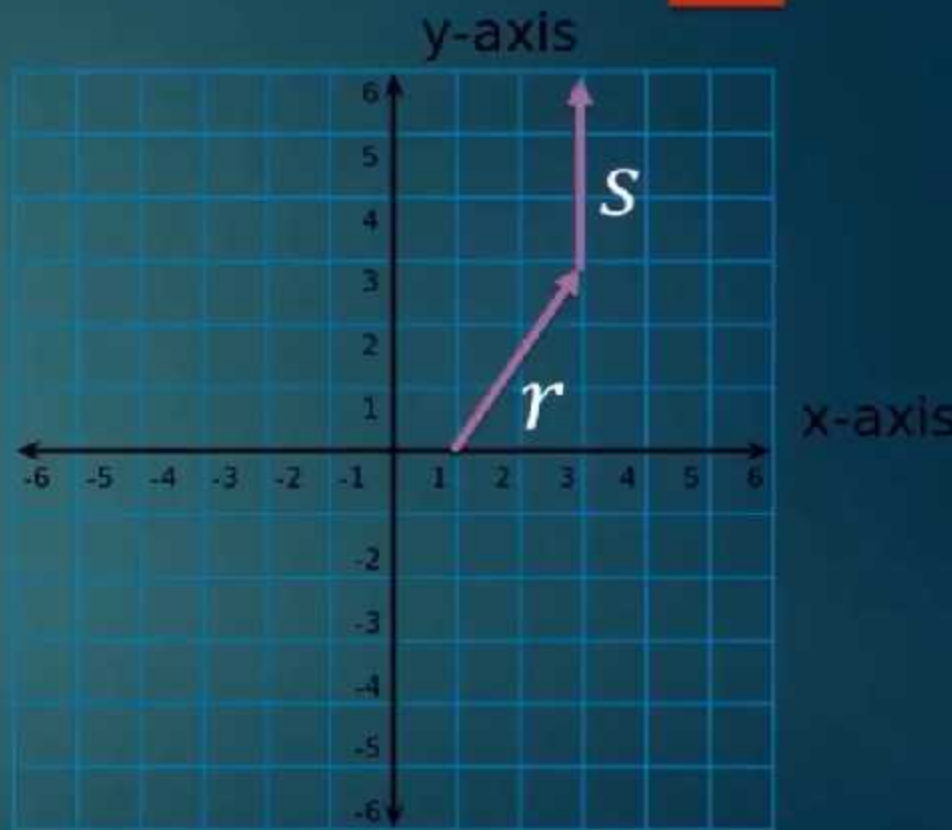
Operations on Vectors

4

► 1- Additions:

$$r = 2i + 3j$$

$$s = 0i + 3j$$



Operations on Vectors

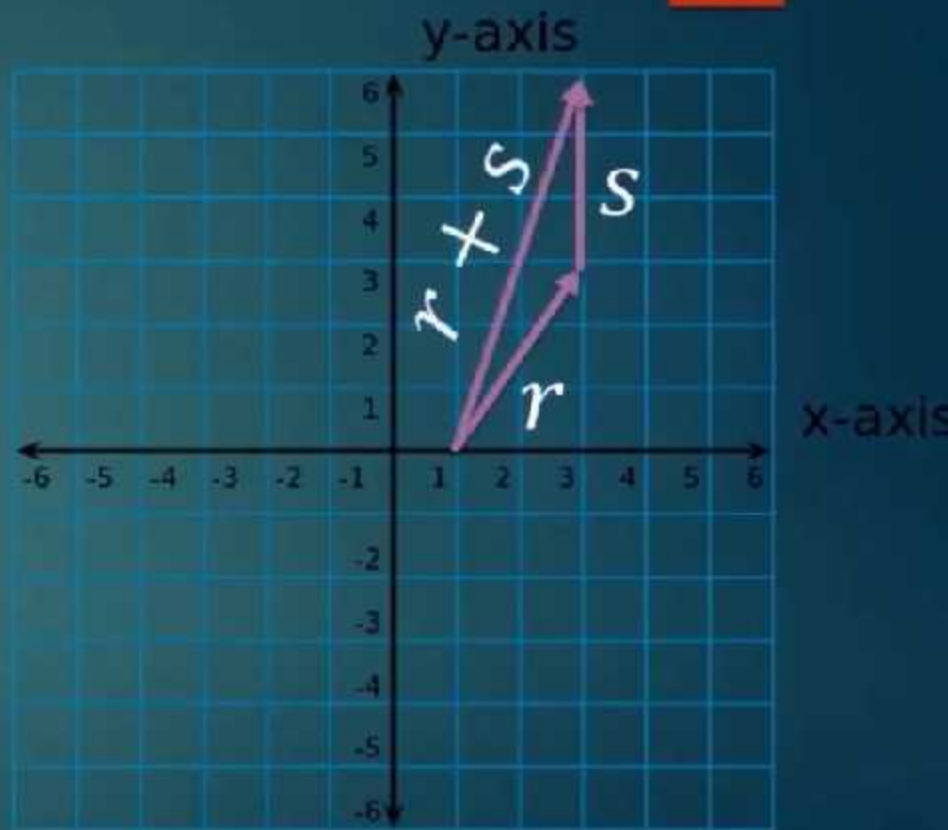
4

► 1- Additions:

$$r = 2i + 3j$$

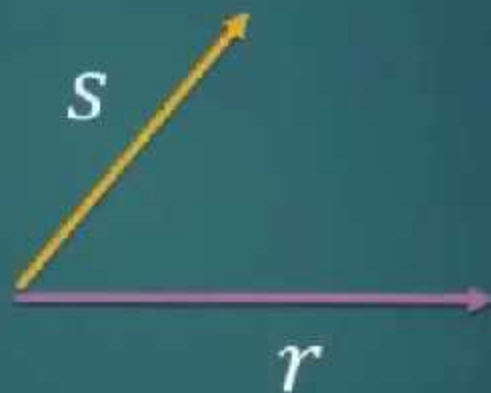
$$s = 0i + 3j$$

$$r + s = 2i + 6j$$



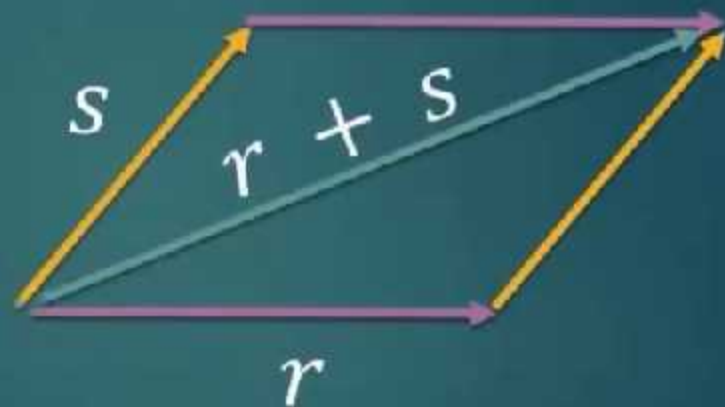
Operations on Vectors

5



Operations on Vectors

5



Operations on Vectors

6

$$r = 2i + 3j$$



Operations on Vectors

6

$$r = 2i + 3j$$



$$2r = 2(2i + 3j) = 4i + 6j$$



Operations on Vectors

6

$$r = 2i + 3j$$


$$2r = 2(2i + 3j) = 4i + 6j$$


$$-r = -(2i + 3j) = -2i - 3j$$

Operations on Vectors

6

$$r = 2i + 3j$$


$$2r = 2(2i + 3j) = 4i + 6j$$


$$-r = -(2i + 3j) = -2i - 3j$$


Operations on Vectors

6

$$r = 2i + 3j$$


$$2r = 2(2i + 3j) = 4i + 6j$$


$$-r = -(2i + 3j) = -2i - 3j$$


$$-2r = -2(2i + 3j) = -4i - 6j$$

Operations on Vectors

6

$$r = 2i + 3j$$


$$2r = 2(2i + 3j) = 4i + 6j$$


$$-r = -(2i + 3j) = -2i - 3j$$


$$-2r = -2(2i + 3j) = -4i - 6j$$


Operations on Vectors

7

Magnitude or Modulus

Angle or Direction

Operations on Vectors

7

Magnitude or Modulus

$$r = 3i + 4j$$

$$||r|| = \sqrt{3^2 + 4^2}$$

$$||r|| = 5$$

Angle or Direction

$$\angle r = \tan^{-1} \frac{4}{3}$$

Operations on Vectors

7

Magnitude or Modulus

$$r = 3i + 4j$$

$$||r|| = \sqrt{3^2 + 4^2}$$

$$||r|| = 5$$

Angle or Direction

$$\angle r = \tan^{-1} \frac{4}{3}$$

$$\angle r = 53.13^\circ$$

Operations on Vectors

7

Magnitude or Modulus

$$r = 3i + 4j$$

$$||r|| = \sqrt{3^2 + 4^2}$$

$$||r|| = 5$$

Angle or Direction

$$\angle r = \tan^{-1} \frac{4}{3}$$

$$\angle r = 53.13$$

Polar Form

$$r = 5 \angle 53.13$$

Operations on Vectors

8

Cartesian Form



Polar Form

$$r = 3i + 4j$$

$$\|r\| = \sqrt{3^2 + 4^2} = 5$$

$$\angle r = \tan^{-1} \frac{4}{3} = 53.13$$

$$r = 5 \angle 53.13$$

Operations on Vectors

Cartesian Form



Polar Form

$$r = -3i + 4j$$

$$\|r\| = \sqrt{3^2 + 4^2} = 5$$

Operations on Vectors

9

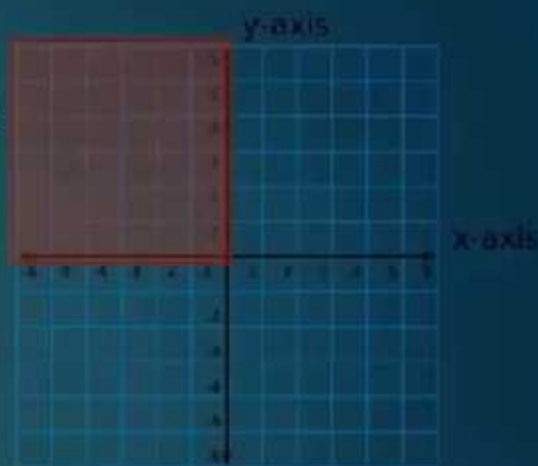
Cartesian Form



Polar Form

$$r = -3i + 4j$$

$$\|r\| = \sqrt{3^2 + 4^2} = 5$$



Operations on Vectors

9

Cartesian Form

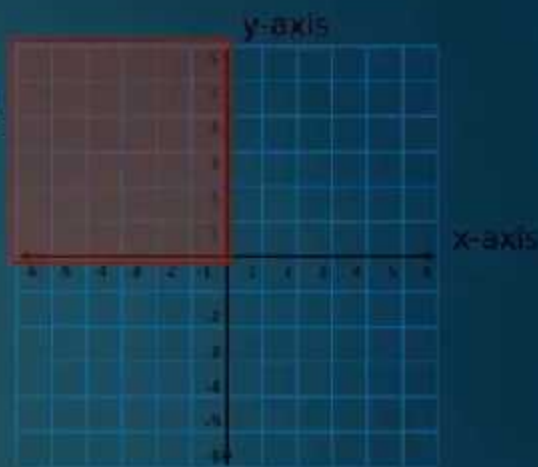


Polar Form

$$r = -3i + 4j$$

$$\|r\| = \sqrt{3^2 + 4^2} = 5$$

$$\angle r = \tan^{-1} \frac{4}{3} = 53.13$$



Operations on Vectors

9

Cartesian Form

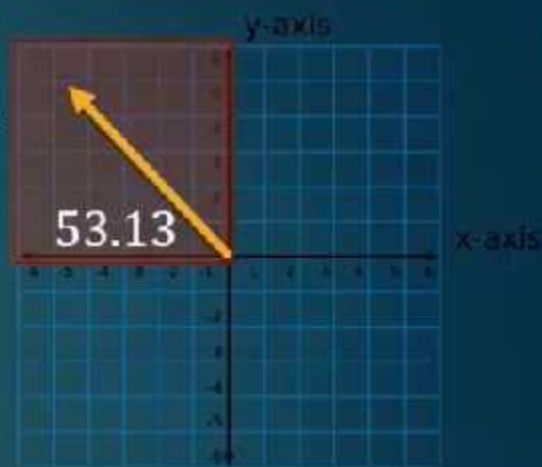


Polar Form

$$r = -3i + 4j$$

$$\|r\| = \sqrt{3^2 + 4^2} = 5$$

$$\angle r = \tan^{-1} \frac{4}{3} = 53.13$$



Operations on Vectors

9

Cartesian Form



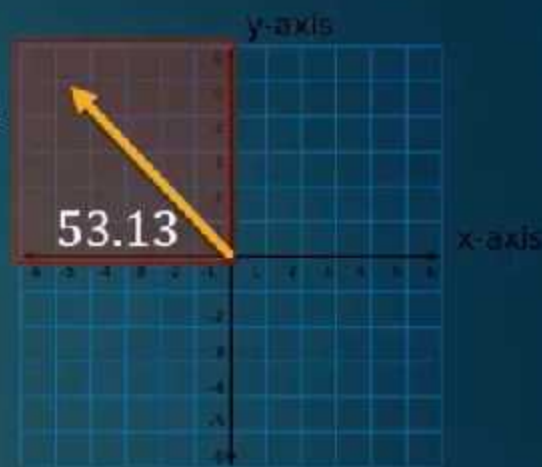
Polar Form

$$r = -3i + 4j$$

$$\|r\| = \sqrt{3^2 + 4^2} = 5$$

$$\angle r = \tan^{-1} \frac{4}{3} = 53.13$$

$$\theta = 180 - 53.13 = 126.87$$



Operations on Vectors

9

Cartesian Form



Polar Form

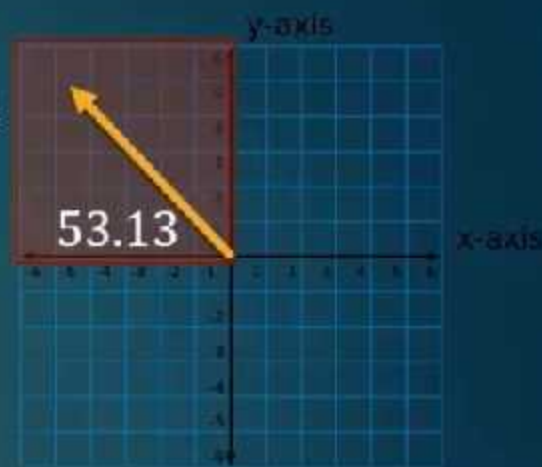
$$r = -3i + 4j$$

$$\|r\| = \sqrt{3^2 + 4^2} = 5$$

$$\angle r = \tan^{-1} \frac{4}{3} = 53.13$$

$$\theta = 180 - 53.13 = 126.87$$

$$r = 5 \angle 126.87$$



Operations on Vectors

10

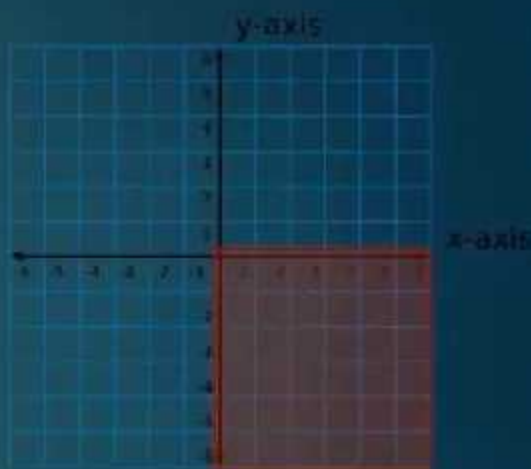
Cartesian Form



Polar Form

$$r = 3i - 4j$$

$$\|r\| = \sqrt{3^2 + 4^2} = 5$$



Operations on Vectors

10

Cartesian Form

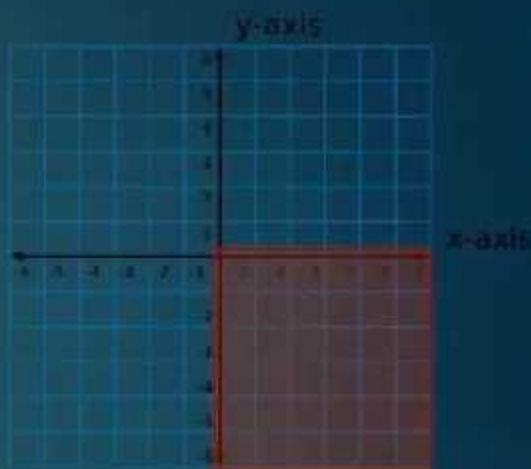


Polar Form

$$r = 3i - 4j$$

$$\|r\| = \sqrt{3^2 + 4^2} = 5$$

$$\angle r = \tan^{-1} \frac{4}{3} = 53.13$$



Operations on Vectors

10

Cartesian Form

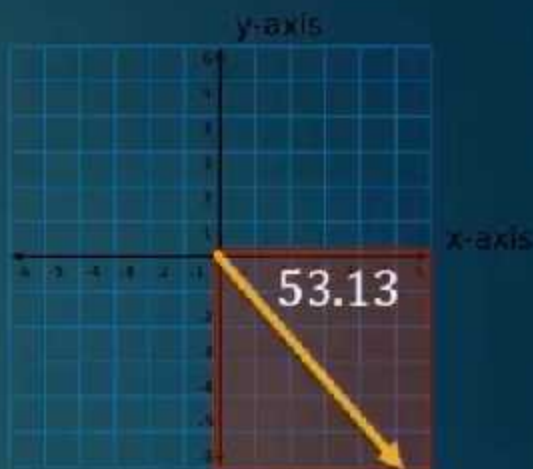


Polar Form

$$r = 3i - 4j$$

$$\|r\| = \sqrt{3^2 + 4^2} = 5$$

$$\angle r = \tan^{-1} \frac{4}{3} = 53.13$$



Operations on Vectors

10

Cartesian Form



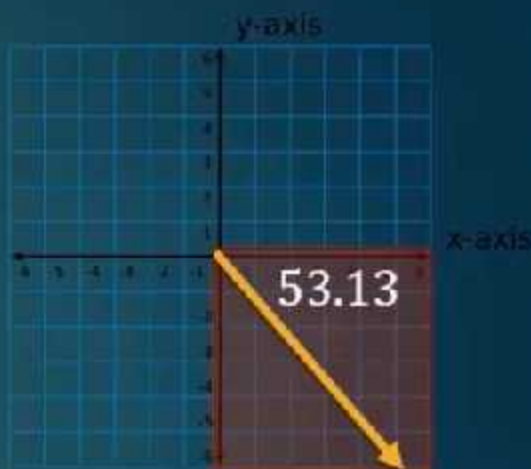
Polar Form

$$r = 3i - 4j$$

$$\|r\| = \sqrt{3^2 + 4^2} = 5$$

$$\angle r = \tan^{-1} \frac{4}{3} = 53.13$$

$$\theta = 360 - 53.13 = 306.87$$



Operations on Vectors

10

Cartesian Form



Polar Form

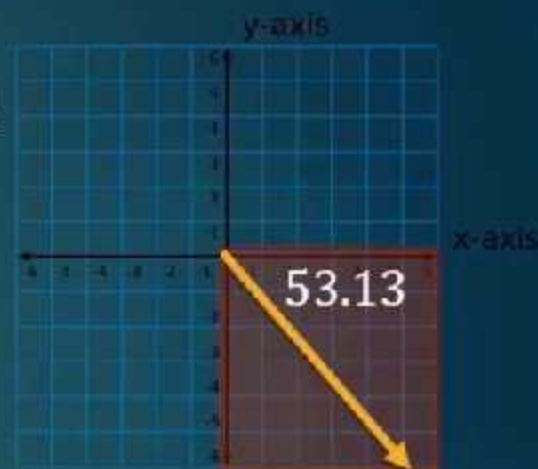
$$r = 3i - 4j$$

$$\|r\| = \sqrt{3^2 + 4^2} = 5$$

$$\angle r = \tan^{-1} \frac{4}{3} = 53.13$$

$$\theta = 360 - 53.13 = 306.87$$

$$r = 5 \angle 306.87$$



Operations on Vectors

11

Cartesian Form



Polar Form

$$r = -3i - 4j$$

$$||r|| = \sqrt{3^2 + 4^2} = 5$$

Operations on Vectors

11

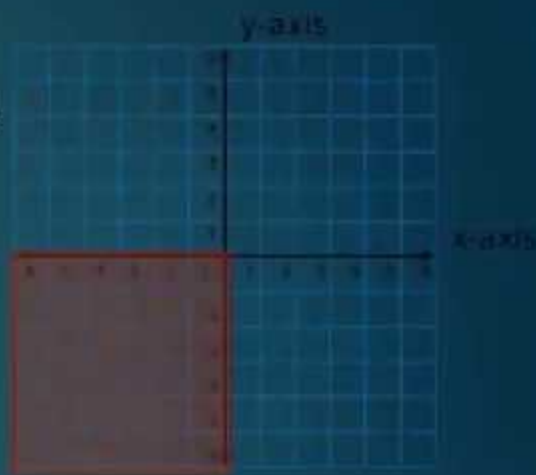
Cartesian Form



Polar Form

$$r = -3i - 4j$$

$$\|r\| = \sqrt{3^2 + 4^2} = 5$$



Operations on Vectors

11

Cartesian Form

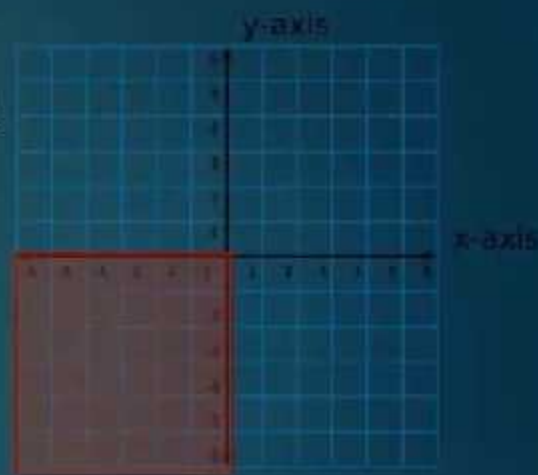


Polar Form

$$r = -3i - 4j$$

$$||r|| = \sqrt{3^2 + 4^2} = 5$$

$$\angle r = \tan^{-1} \frac{4}{3} = 53.13$$



Operations on Vectors

11

Cartesian Form

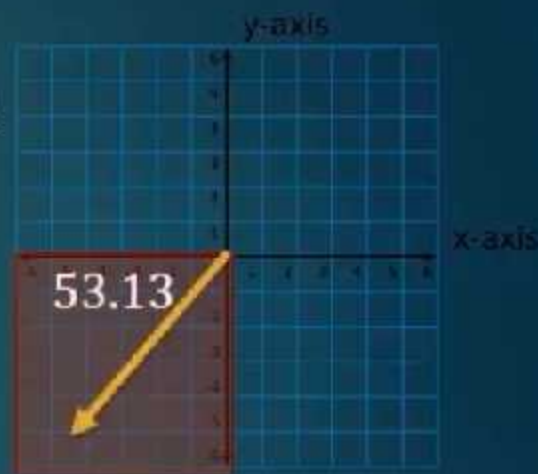


Polar Form

$$r = -3i - 4j$$

$$\|r\| = \sqrt{3^2 + 4^2} = 5$$

$$\angle r = \tan^{-1} \frac{4}{3} = 53.13$$



Operations on Vectors

11

Cartesian Form



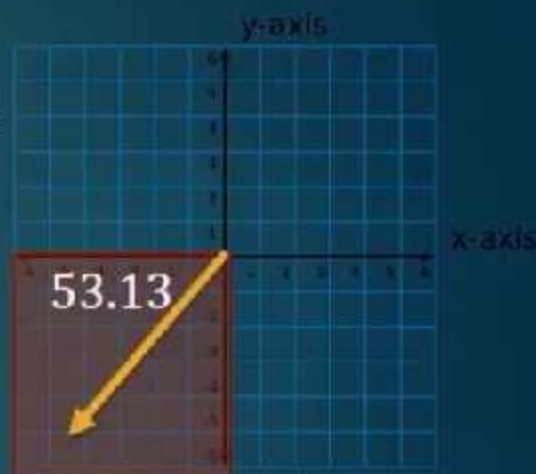
Polar Form

$$r = -3i - 4j$$

$$\|r\| = \sqrt{3^2 + 4^2} = 5$$

$$\angle r = \tan^{-1} \frac{4}{3} = 53.13$$

$$\theta = 180 + 53.13 = 233.13$$



Operations on Vectors

11

Cartesian Form



Polar Form

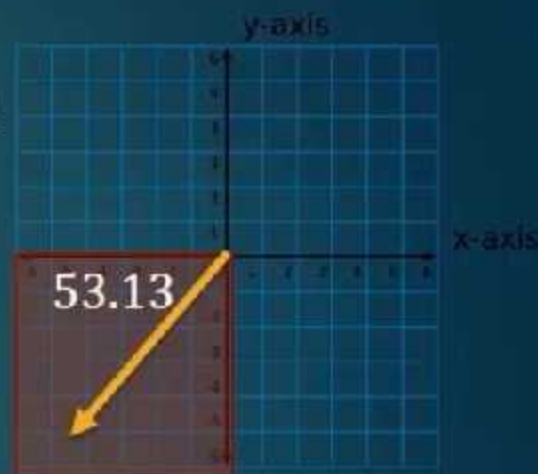
$$r = -3i - 4j$$

$$\|r\| = \sqrt{3^2 + 4^2} = 5$$

$$\angle r = \tan^{-1} \frac{4}{3} = 53.13$$

$$\theta = 180 + 53.13 = 233.13$$

$$r = 5 \angle 233.13$$



Operations on Vectors

12

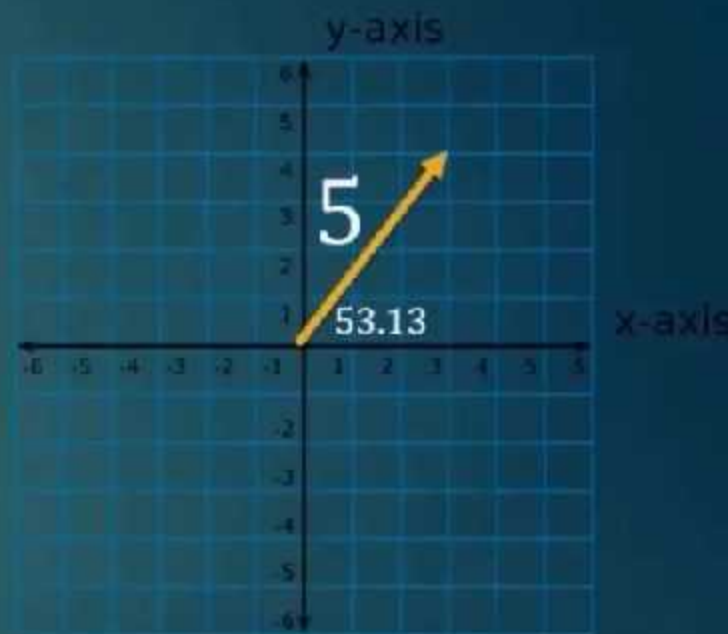
Polar Form  Cartesian Form

Operations on Vectors

12

Polar Form $\longrightarrow$ Cartesian Form

$$r = 5 \angle 53.13$$



Operations on Vectors

12

Polar Form $\longrightarrow$ Cartesian Form

$$r = 5 \angle 53.13$$



Operations on Vectors

12

Polar Form $\longrightarrow$ Cartesian Form

$$r = 5 \angle 53.13$$

$$r = 5 \cos 53.13 \, i + 5 \sin 53.13 \, j$$

